Jigyasa Verma

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Summary

I'm an experimentalist who builds wet-lab methods that generate data quickly, accessibly, and reproducibly, and I use that data to check whether AI-designed molecules behave as predicted. I have moved across yeast genetics, mammalian cell biology, and RNA biochemistry, picking up each new system quickly and working closely with computational and Eterna design teams whose molecules I test. I ran robotic genome-wide screens and set up automated live-cell imaging on my own, and I build protocols that others can run: I invented FAST-MaP, wrote it up as an invited step-by-step protocol, and transferred it to AI@HHMI. Turning expert science into procedures that others can execute reliably is the work I most enjoy, and automated labs are where I want to take it next.

Research Experience

Postdoctoral Researcher Mar 2025 – Present
Stanford University, Das Lab *California, USA*

- **Invented FAST-MaP, wrote it up as an invited step-by-step protocol, and transferred it to AI@HHMI**; the primer-less sequencing pipeline takes a DNA template to per-nucleotide RNA reactivity in about one week (∼24–30 h hands-on) using commercial gene synthesis and Nanopore sequencing services, standard bench equipment, and the public `cmuts` web server, with no in-house sequencing or coding.
- **Designed a 50-nt anti-5′UTR in the 3′UTR** to pair with the 5′UTR of the same mRNA, which represses cap-dependent NanoLuc translation in HeLa lysate while a scrambled control does not (unpublished).
- **Designed and launched *mRNA Switch Controls***, an Eterna crowdsourcing lab in which players design 50-nt anti-5′UTR sequences in the 3′UTR to switch NanoLuc translation off or on; selected pilot-round designs are in validation.
- **Invented the method (FAST-MaP) that generated chemical-mapping data for a deep-learning RNA design study**, showing how the starting sequence, the top Eterna design, and an AI (MPNN-fixbb) design of a c-di-GMP riboswitch change shape when the ligand binds (*Science*, 2026).

Postdoctoral Researcher Jan 2024 – Aug 2024
University of British Columbia, Loewen and Roskelley Labs *Vancouver, Canada*

- **Performed quantitative image analysis** of microscopy data for cellular phenotyping.

Graduate Research Assistant Sep 2017 – Dec 2023
University of British Columbia, Loewen and Roskelley Labs *Vancouver, Canada*

- **Ran genome-wide chemical-genetic and *pan2*Δ synthetic genetic array screens** (∼4,800 strains under nocodazole, 1536-colony format on a Singer RoToR, scored in Balony), which, with live-cell imaging of yeast and PAN2 shRNA knockdown in HEK293A cells, showed that PAN2–PAN3 deadenylase activity is required for mitotic robustness under microtubule stress from yeast to human (first author, in review).
- **Validated screen and RNA-seq hits with liquid growth assays, RT-qPCR, and genetic analysis**, which showed that catalytic-dead Pan2 (D1020A) fails to rescue nocodazole sensitivity, human PAN2 rescues only when Pan3 is present, and *CLB1* and *CLN3* shift in the direction RNA-seq predicted.
- **Independently set up automated live-cell imaging on the Zeiss Celldiscoverer 7**, teaching myself the instrument and building the 96-well workflow behind the yeast time-lapse data in the PAN2 manuscript.

Graduate Student Researcher Aug 2014 – May 2017
Jawaharlal Nehru Centre for Advanced Scientific Research *Bangalore, India*

- **Screened *atg* deletion mutants and imaged mitotic spindles**, which showed that *atg11*Δ is sensitive to the microtubule drugs thiabendazole and benomyl and has shorter spindles than wild type (co-second author, *PLOS Biology*, 2025).

Publications

[**Published**] Townley J.*, Kladwang W.*, … **Verma J.**, … Eterna Participants, Das R. De novo design of RNA pseudoknots with deep learning. *Science* (2026). *equal contribution

[**Published**] Reza M.H., Aggarwal R.*, **Verma J.*** et al. Autophagy-related protein Atg11 is essential for microtubule-mediated chromosome segregation. *PLOS Biology* (2025). *equal contribution

[**In review**] **Verma J.** et al. FAST-MaP: Chemical Mapping of RNA Structures Using Primer-less Sequencing. Invited protocol; under review, 2026. Preprint

[**In review**] **Verma J.** et al. PAN2–PAN3 deadenylase activity is selectively required for mitotic robustness under microtubule stress. Under review, 2026. Preprint

Technical Expertise

Lab Automation & Instrumentation: Singer RoToR robotic pinning with Balony colony scoring (genome-wide chemical-genetic and SGA screens); Zeiss Celldiscoverer 7 automated live-cell imaging (yeast, 96-well)

RNA Biology & Structure Probing: *in vitro* chemical probing (2A3, DMS) with mutational-profiling readout; *in vitro* transcription; mRNA construct design and NanoLuc translation reporter assays (HeLa lysate); sample preparation for commercial Nanopore sequencing; RNA and DNA QC (Qubit, agarose gels); RT-qPCR

Molecular & Cellular Biology: cloning and site-directed mutagenesis; shRNA knockdown (lipofectamine transfection); western blot; MTT viability assays; HEK293A culture; yeast genetics (PCR-based genomic modification, SGA mating and selection, spot and growth assays)

Imaging & Analysis: live-cell time-lapse microscopy; quantitative image analysis (Fiji; custom Python segmentation pipelines built with AI coding assistants); chemical-mapping analysis (`cmuts`)

Awards

mRNA Research Advancement Award, CELLSCRIPT (2026)

mRNA Innovators Award, Eclipsebio (2025)

President's Academic Excellence Initiative PhD Award, University of British Columbia (2021–2023)

Raymond A. Pederson Prize in Physiology, University of British Columbia (2020)

Teaching and Service

Peer Reviewer: *Science*, *Nature Communications*, *Nature Nanotechnology*, *Chemistry – A European Journal* (2025–2026)

Selected Presentations: Stanford RNA Medicine Symposium (poster, 2026); Stanford Biochemistry Retreat (poster, 2026); Salk Cell Cycle Meeting (talk, 2023); Gordon Research Conference on Centromeres in Health & Disease (talk, 2022)

Mentorship: trained undergraduates and junior researchers in RNA biochemistry, cloning, and imaging

Education

PhD, Cell and Developmental Biology, University of British Columbia, Vancouver	2017–2023
MS, Molecular Biology and Genetics, JNCASR, Bangalore	2014–2017
BSc, Microbiology, University of Delhi	2011–2014